V.L Memorial Public School

475 Vijay Nagar

CHAPTER-13-QUESTIONS-CLASS8TH-MATHS-REPEAT

Class 08 - Maths (Ganita Prakash)

Section A

1. **Fill in the blanks:** **[6]**

   (a) The bottom row of a pyramid is 2, 4, 6. **[1]**
   The number at the top is _____.

   (b) Using the digits 2, 6 and 9, the largest possible product is _____. **[1]**

   (c) After performing the steps multiply by 2, add 6, divide by 2 and subtract the original number, the final result is _____. **[1]**

   (d) If the bottom row of a 3-row pyramid is a, b, c, the number at the top is _____. **[1]**

   (e) If a trick ends with the expression x + 7 - x, the final answer is _____. **[1]**

   (f) The value of x in $\frac{1}{3} + x = \frac{2}{5}$ is _____. **[1]**

Section B

2. Bottom row of a pyramid is: 4, 1, 6 **[1]**
   What is the top number?

   a) 12 b) 14

   c) 11 d) 13

3. If the top-left number of a 2 × 2 calendar grid is 11, what is the bottom-right number? **[1]**

   a) 18 b) 17

   c) 19 d) 20

4. A number is formed as abcabc. It is divisible by: **[1]**

   a) 11 only b) 7 only

   c) 13 only d) 7, 11 and 13

5. A student performs these operations on a number x: **[1]**
   Multiply by 4, add 12, then divide by 4, then subtract the original number.
   What is the final answer?

   a) 2 b) 3

   c) 1 d) 4

6. Make the largest product possible using digits 4, 7, and 3. **[1]**

   a) 301 b) 305

   c) 141 d) 304

7. Let a two-digit number be 10a + b. **[1]**
   The difference between the number and its reverse is:

a) 9(b - a)

b) 9(a + b)

c) 11(a - b)

d) 10(b - a)

8. For any three-digit number abc, the sum of abc + bca + cab is always divisible by: **[1]**

a) 37

b) 8 only

c) 9 only

d) 11 only

9. Which of the following expressions will always give the same value for any starting number x? **[1]**

a) $\frac{x}{2} + 1$

b) $x^2 - x$

c) $2x + 5$

d) $5x - 5x + 4$

10. Using digits 3, 5 and 9, the largest product is: **[1]**

a) 59 × 3

b) 95 × 3

c) 53 × 9

d) 35 × 9

11. For a 3-row pyramid with bottom entries a, b, c, the top number is: **[1]**

a) $a + 2b + c$

b) $2a + b + 2c$

c) $a + 3b + c$

d) $a + b + c$

12. For what value of x , a number of the form $(x \times 100 + 6 \times 10 + 8)$ is divisible by 3? **[1]**

a) 9

b) 8

c) 5

d) 7

13. Find the sum of a 2×2 grid, if the top left number of the grid is 11. **[1]**

a) 66

b) 61

c) 54

d) 60

14. Rohan uses a trick to find a friend's birthday. He tells his friend to multiply the month number M by 5, add 7, multiply by 4, add 13, and finally multiply the result by 5. Which algebraic expression represents this trick? **[1]**

a) $5M + 7 + 4 + 13 + 5$

b) $(5M + 7) \times 4 + 13 \times 5$

c) $\{(5M + 7) \times 4 + 13\} \times 5$

d) $5(M + 7 + 4 + 13)$

15. A number trick starts with a number x. After some steps, the final expression becomes **[1]**
3x + 7 - 3x
What will be the final result?

a) x

b) 3

c) 3x

d) 7

16. 37037 is always divisible by **[1]**

a) 307

b) 31

c) 101

d) 17

17. If three digits are p < q < r, the arrangement that gives the largest product is: **[1]**

a) rp × q

b) pq × r

c) rq $\times$ p

d) qp $\times$ r

18. Using digits 1, 3 and 7 once each, the largest possible product is: **[1]**

a) 13 × 7

b) 17 × 3

c) 31 × 7

d) 71 × 3

19. Think of a date $\rightarrow$ Add 10 to the month $\rightarrow$ multiply by 5 $\rightarrow$ subtract 40 from it $\rightarrow$ multiply by 4 $\rightarrow$ again subtract it by 40 $\rightarrow$ multiply it by 5. **[1]**

What is the constant obtained in the above trick?

a) None of these

b) 240

c) 200

d) 180

20. The number 1001 is not divisible by **[1]**

a) 11

b) 13

c) 17

d) 1

21. In a calendar, the top-left number of a 2×2 block is a. What is the sum of the block? **[1]**

a) $4a + 12$

b) $4a + 8$

c) $4a + 16$

d) $4a + 14$

Section C

22. **Assertion (A):** The sum of the numbers in a '2×2' calendar grid is 100. The top-left number is 21. **[1]**

Reason (R): The sum of numbers in a '2×2' calendar grid is always '$4a + 16$', where 'a' is the top-left number.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

23. **Assertion (A):** For any two-digit number, the difference between the number and its reverse is divisible by 11. **[1]**

Reason (R): If the number is 10a + b, then the difference with its reverse is 9(b - a).

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. **Assertion (A):** If the final expression of a number trick is $x + 5 - x$, the result is always 5. **[1]**

Reason (R): In a valid number trick, the variable must cancel out in the final expression.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. **Assertion (A):** In a 2×2 calendar grid, if the top-left number is a, the sum of the four numbers is 4a + 16. **[1]**

Reason (R): The four numbers in the grid are a, a + 1, a + 7, a + 8.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** If you think of a number, double it, add 6, divide by 2, and subtract the original number, the result is always 3. **[1]**

Reason (R): The algebraic expression for the trick simplifies to $(2x + 6) \div 2 - x = 3$.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** If the bottom row of a 3-row number pyramid is a, b, c then the top number is a + 2b + c. **[1]**

Reason (R): In a number pyramid, each number is the product of the two numbers directly below it.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. **Assertion (A):** Using any three digits, the largest product is always obtained by using the largest digit as the multiplier and arranging the other two digits in decreasing order. **[1]**

Reason (R): Placing the largest digit in the tens place always gives the largest product.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

Section D

29. A number is multiplied by 3, then 12 is added, the result is divided by 3 and finally the original number is subtracted. Find the final result. **[1]**

30. Using the digits 4, 6 and 8 only once, form the largest possible product. **[1]**

31. Explain why the trick “multiply a number by 4 and subtract four times the original number” always gives 0. **[1]**

32. Find the top number if the bottom row is 2, 5, 4 **[1]**

33. Show that the number formed as abcabc is divisible by 7, 11 and 13. **[1]**

34. In a 2 × 2 calendar grid, if the top-left number is a, write the bottom-right number. **[1]**

35. Write the expression for the top of a 3-row pyramid if the bottom row is a, b, c. **[1]**

36. If the final step of a trick gives the expression x + 11 - x, what is the result? **[1]**

37. The sum of a 2 × 2 calendar block is 44. Find the top-left number. **[1]**

38. State the rule for forming the largest product using three digits. **[1]**

39. In a date trick where you multiply month M by 5, add 6, multiply by 4, add 9, multiply by 5, and add day D, what number must you subtract from the final answer to decode the date? **[2]**

40. Sumit is three years older than Meenakshi. The sum of their ages is 39. Find the age of each. **[2]**

41. A father says to his son, "In 5 years, my age will be three times your age at that time." If the father's present age is 5 years more than 4 times his son's age, find the present age of the son. **[2]**

42. Write an expression for the top of a pyramid if the base has a, b, c, and d terms. **[2]**

43. In a calendar, Meera selected a 2 × 2 grid of dates. The sum of all four numbers in the grid was 52. Find the four dates she selected. **[2]**

44. One-fourth of the birds of a flock are at a riverbank, and one-fifth of that flock are in their nests. The remaining 22 birds are wandering in search of food. What is the number of birds that are in their nest? **[2]**

45. Using the date trick, someone follows the steps and gets 894 as the final answer. What date did they start with? **[2]**

46. Start with a number, triple it, add 12, divide by 3, subtract the original number. What is the final result? Prove it algebraically. **[2]**

47. Write a general expression for the topmost number of a pyramid with 5 rows whose bottom row is a, b, c, d, and e. **[2]**

48. Two cowherds, Gauri and Naina, meet on the road. Gauri says Naina has twice as many cows. Naina says if she gave 3 cows to Gauri, they would have equal cows. How many cows do each have? **[2]**

49. Consider a pyramid where the bottom row is: 2, 4, 6, 8 **[3]**

    i. Complete the entire pyramid.

    ii. Find a general formula for the top number if the bottom row is: a, a + d, a + 2d, a + 3d

    iii. Verify your formula with the pyramid in part (i).

50. Modified Largest Product: **[3]**

    You have four digits: 2, 3, 5, 7. Arrange them to form two 2-digit numbers and multiply them.

    i. What arrangement gives the largest product?

    ii. What arrangement gives the smallest product?

    iii. Prove your answers using systematic analysis.

51. Three-digit cycling problem: **[3]**

    Take a 3-digit number abc (where a, b, c are digits). Create two more numbers by cycling: bca and cab.

    i. Prove that the sum abc + bca + cab is always divisible by 37.

    ii. Prove that this sum is also always divisible by 3.

    iii. Test with the number 234.

52. In Harshit and the Genie problem: **[3]**

    i. If after going around the tree 3 times and paying the genie 8 coins each time, Harshit is left with 0 coins, how many coins did he start with?

    ii. Verify your answer by working through the rounds.

    iii. If the genie charges 5 coins per round instead of 8, how many coins would Harshit need to start with to end with 0 coins after 3 rounds?

53. Create your own variation by changing the multipliers and additions. Design a trick where you multiply the month by 4, add 5, multiply by 5, add 7, multiply by 4, and add the day. **[3]**

    i. What expression do you get for the final answer?

    ii. What number must you subtract to decode the date?

    iii. Test your trick with the date 15/03 (March 15).

54. A devotee starts with some flowers and visits 4 shrines. At each shrine's pond, his flowers double. After each doubling, he offers some flowers at the shrine. He offers the same number of flowers at all 4 shrines and is left with zero flowers at the end. What is the minimum number of flowers the devotee could have been carrying at the start? **[3]**

55. Find the dates if the final answers are the following: **[3]**

i. 1269

ii. 394

iii. 296

56. Calendar Grid with Different Shape: **[3]**

Consider a calendar. Instead of a 2 × 2 grid, you pick an L-shaped region consisting of 3 consecutive dates in one row.

i. If the smallest number in the L-shape is a, express all three numbers in terms of a.

ii. If the sum of the three numbers is 36, find the three dates.

iii. Create a similar trick for a 3 × 1 vertical strip.

57. In a number pyramid with 3 rows, if the bottom row contains a, b, c, write an expression for the top number. Use this to find the top number when the bottom row is 4, 7, 5. **[3]**

58. Create your own 'Think of a Number' trick that results in the original number. **[3]**

i. Design the trick with at least 5 steps.

ii. Prove algebraically that it works.

iii. Test it with three different starting numbers.

59. In the date trick, Mukta chose 26/01 (Republic Day) and followed these steps: **[5]**

i. Multiply the month by 5

ii. Add 6

iii. Multiply by 4

iv. Add 9

v. Multiply by 5

vi. Add the day

Her final answer was 291. Explain using algebra how Shubham figured out the date.

60. A person picks a 2-digit number with different digits, reverses it, finds the difference, and divides by 9. Using algebra, prove there will be no remainder and find the relationship between the quotient and the original digits. **[5]**

61. Explain how algebra helps us understand 'Think of a Number' tricks. Use the following trick as an example: **[5]**

i. Think of a number

ii. Double it

iii. Add four

iv. Divide by two

v. Subtract the original number

62. A number pyramid has 4 rows. The bottom row contains four consecutive odd numbers starting from 1. **[5]**

i. Fill in the complete pyramid

ii. Write an expression for the top number in terms of a, b, c, d (bottom row values)

iii. Verify your expression with the pyramid from part (i)

iv. If the first four Virahāṅka-Fibonacci numbers (1, 2, 3, 5) are in the bottom row, what number appears at the top? Are all numbers in the pyramid Virahāṅka-Fibonacci numbers?

63. Consider any 3-digit number abc. Make two other 3-digit numbers by cycling the digits to get bca and cab. Add all three numbers. Using algebra, prove the sum is always divisible by 37 and also by 3. **[5]**

64. Explain how to find the largest product when filling three digits (2, 3, and 5) in the format AB × C. Use both systematic checking and algebraic proof. **[5]**

65. Karim goes around a magical tree 3 times. Each time: **[5]**

The money in his pocket doubles

He gives 8 coins to the genie

After the third round, he has exactly 8 coins left (which he gives to the genie, leaving him with 0).

i. How many coins did Karim initially have?

ii. Draw a diagram showing the number of coins after each step

iii. If Karim wants to have coins left after 3 rounds, what should be the maximum cost per round he agrees to?

iv. Write an algebraic expression for the number of coins Karim has after n rounds

v. If the genie wants all of Karim's coins after 3 rounds, and Karim starts with 20 coins, what should be the cost per round?

66. For a 3-row number pyramid with bottom row a, b, c, derive the expression for the top number. **[5]**

Use this to find the top number if the bottom row is 5, 8, 3.2

Section E

67. **Read the following text carefully and answer the questions that follow:** **[4]**

Shubham then showed Riya a date prediction trick. He asked her to think of a date (month and day). Then he gave these instructions: Multiply the month by 5. Add 6. Multiply by 4. Add 9. Multiply by 5. Add the day. Riya followed the steps and got 291. Shubham immediately said, "You thought of 26th January!" Riya was surprised. Shubham explained that by letting M be the month and D be the day, the final expression becomes 100M + 165 + D. Subtracting 165 gives 100M + D, where the last two digits are the day and the remaining digits are the month. Riya tried another date and got 1390, and Shubham correctly said it was 25th December.

Questions:

i. If the month is March (3) and the day is 10, what will be the answer after the first step (multiply month by 5)? **(1)**

ii. If someone follows all steps and gets 394 as their final answer, what will be the result after subtracting 165? **(1)**

iii. Riya thinks of a date and gets 1269 as her final answer. What date did she think of? Show your work. **(2)**

OR

Explain in simple words how Shubham finds the date from the final answer. **(2)**

68. **Read the following text carefully and answer the questions that follow:** **[4]**

Riya and her friends love playing with numbers. One day, Shubham showed them a 'think of a number' trick. He asked Riya to think of a number, double it, add four, divide by two, and subtract the original number. Riya got 2 every time, no matter what number she started with. Shubham explained that algebra helps us understand why this happens. When we let the unknown number be x and follow the steps, the x cancels out and we are left with 2. Riya was amazed and wanted to learn more such tricks. She tried the trick with different numbers like 5, 10, and 15, and indeed got 2 each time. She realized that algebra is not just about solving equations but also about understanding and creating fun puzzles.

Questions:

i. If the number thought of is 9, what will be the result after step 2 (double it)? **(1)**

ii. If the number thought of is 12, what will be the final answer after all steps? **(1)**

iii. Modify the trick so that the final answer is always 3. Write the new set of instructions clearly. **(2)**

OR

Test your modified trick with the number 8. Show all steps and verify that you get 3. **(2)**

69. **Read the following text carefully and answer the questions that follow:** **[4]**

Shubham also taught Riya about number pyramids. In a number pyramid, each number is the sum of the two numbers directly below it. He showed her a pyramid with three rows. If the bottom row has numbers a, b, and c, then the middle row has a + b and b + c, and the top row has (a + b) + (b + c) = a + 2b + c. Riya practiced with different pyramids. She was given a pyramid where the top number was 60, the middle left number was 12, and the bottom right number was 8. She used algebra to find all the missing numbers.

Questions:

i. In a pyramid, if the bottom row has numbers 3, 6, and 2, what will be the top number? **(1)**

ii. In a pyramid, if the middle row has numbers 14 and 10, what is the top number? **(1)**

iii. In a pyramid with top = 60, middle left = 12, and bottom right = 8, find the missing bottom numbers. Show all steps. **(2)**

OR

Draw the complete pyramid with all numbers filled in. **(2)**